

A Crash Course in Chemical Kinetics



THE BASICS OF REACTION RATE

What is Reaction Rate?

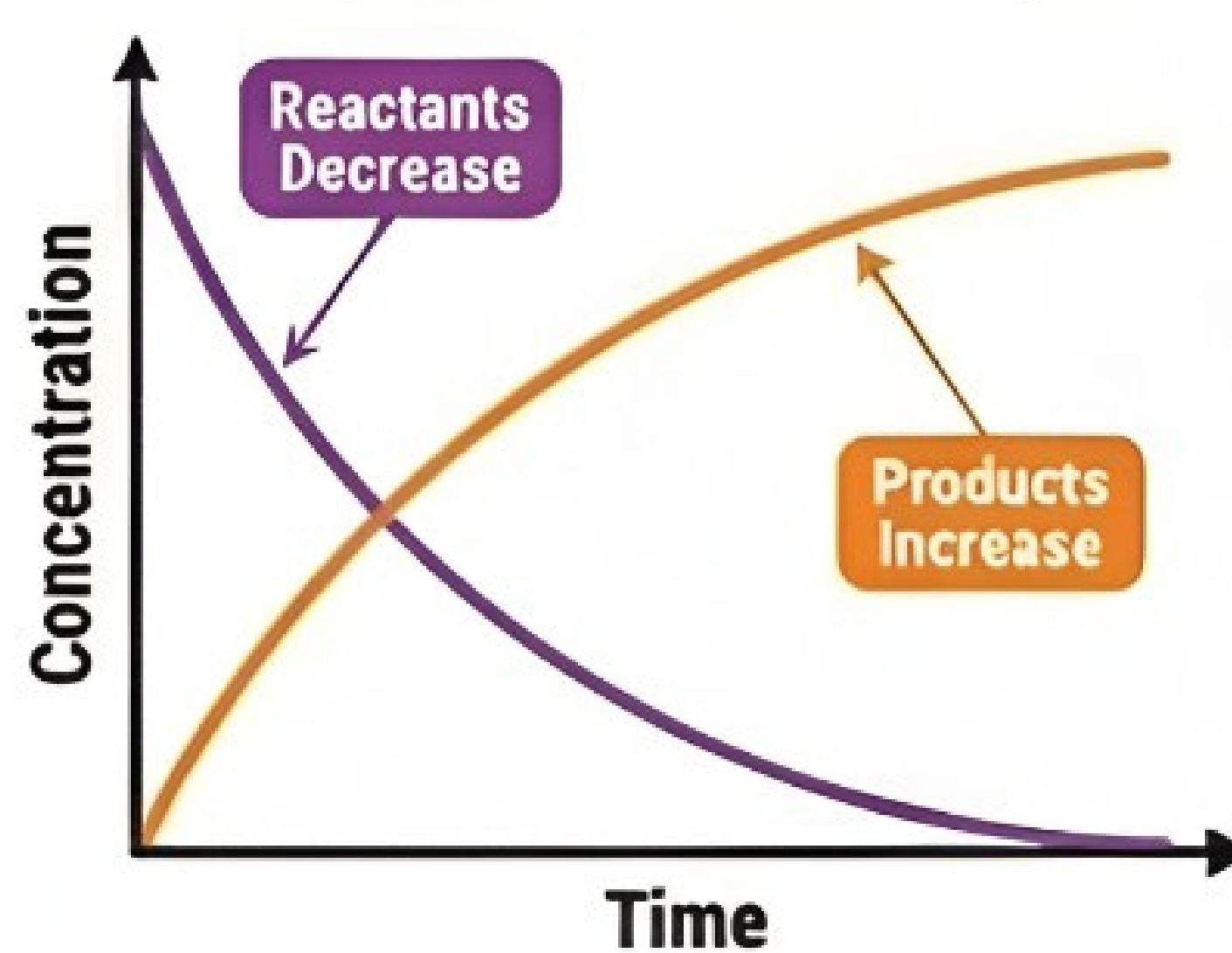
It measures the speed of a reaction, defined as the change in the concentration of a reactant or product over a specific period of time.

General Formula for Rate of Reaction:

$$\text{Rate} = \frac{\text{Change in Concentration}}{\text{Change in Time}}$$

A larger change in concentration over a short time means a faster reaction.

Concentration vs. Time Graphs



Reactants Decrease, Products Increase:
The concentration of reactants decreases over time, while the concentration of products increases. This is a fundamental concept for understanding reaction progress.

EXPRESSING THE RATE OF REACTION

General Chemical Equation



Rate of Disappearance (Reactants)

For A:

$$\text{Rate} = -\frac{1}{3} \cdot \frac{d[A]}{dt}$$

Negative sign indicates a decrease.

Rate of Appearance (Products)

For C:

$$\text{Rate} = +\frac{1}{5} \cdot \frac{d[C]}{dt}$$

Positive sign indicates an increase.

All Rate Expressions are Equal:

For the reaction above,

$$-(1/3) \frac{d[A]}{dt} = -(11) \frac{d[B]}{dt} = +(1/5) \frac{d[C]}{dt}$$

This equality is crucial for solving numerical problems.



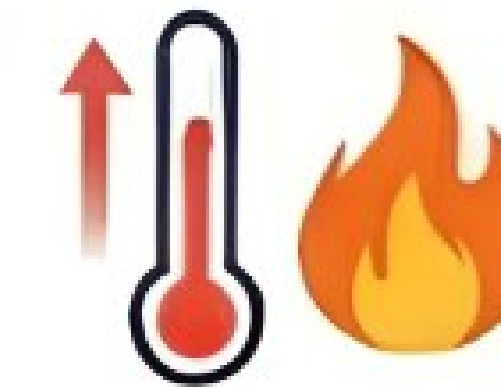
Average vs. Instantaneous Rate

Average Rate uses Δ (D) for a time interval x time interval.



Instantaneous Rate uses the derivative (R) for a specific moment in time.

FACTORS AFFECTING REACTION RATE



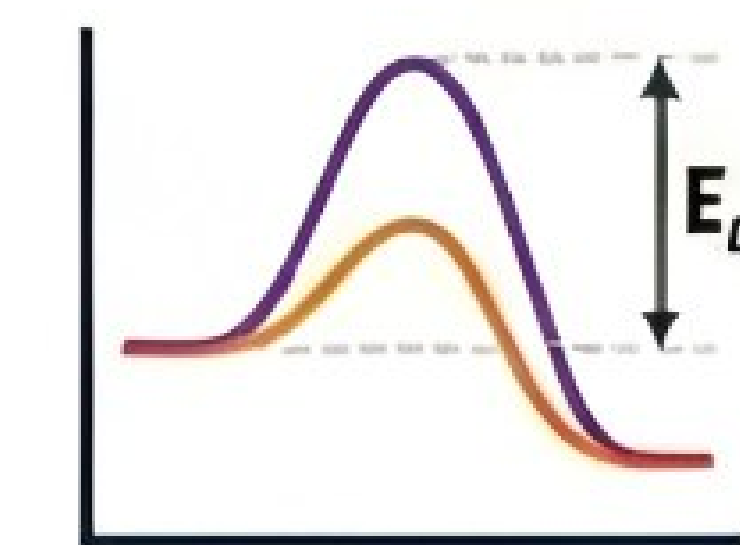
Temperature's Strong Influence

Increasing temperature significantly increases the reaction rate. Rate of thumb: rate approximately doubles for every 10°C rise.

The Arrhenius Equation:

$$k = A \cdot e^{-E_a/RT}$$

Relates rate constant (k) to temperature (T) and activation energy (E_a).



How Catalysts Work

A catalyst increases the reaction rate by providing an alternative reaction pathway with a lower activation energy (E_a).



Collision Theory

For a reaction to occur, reactant molecules must collide with both sufficient energy (equal to or greater than activation energy) and the correct orientation.

RATE LAW & REACTION ORDER

The Rate Law Expression:

An equation that shows how the instantaneous rate depends on the concentration of reactants only.

Rate Laws in Experiments:

The order of a reaction must be determined through experiments, NOT the stoichiometric coefficient.

$$\text{General Rate Law: } \text{Rate} = k \cdot [\text{Reactant}]^n$$

Unit of Rate Constant (k):
 $(\text{mol})^{(1-n)} \cdot (\text{L})^{(n-1)} \cdot \text{s}^{-1}$

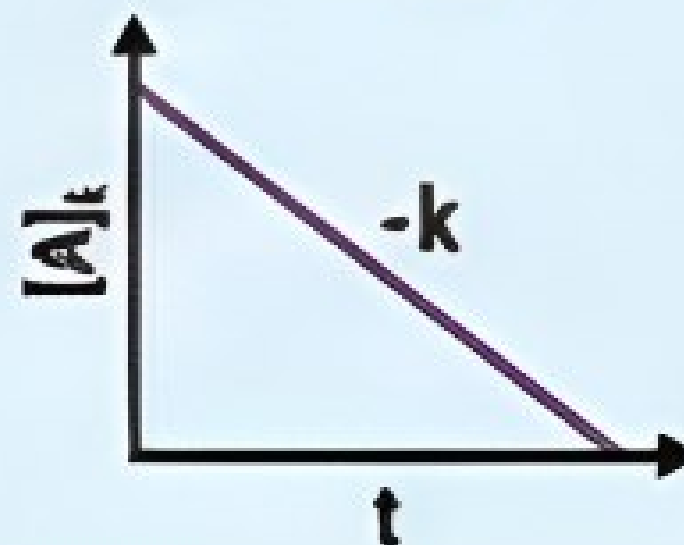
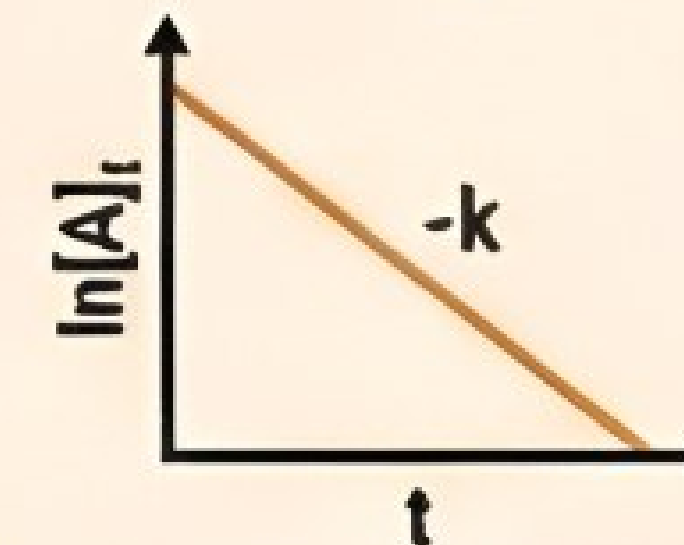
Order "n" is Experimental:

The order of a reaction (n) must be determined through experiments. It is NOT the stoichiometric coefficient.

Properties of Reaction Order (n):

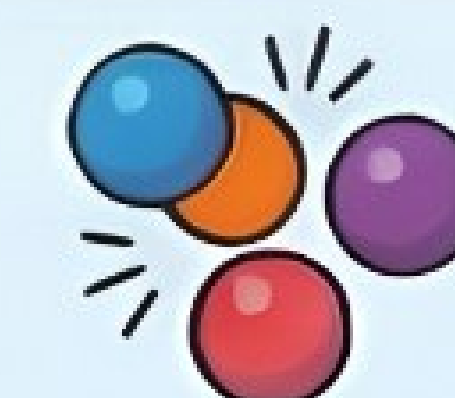
The order can be a zero, integer, or fraction. For multi-reaction expressions, the overall order is the sum of the individual powers.

INTEGRATED RATE LAWS: ZERO VS. FIRST ORDER

Zero-Order Reaction	First-Order Reaction
Rate Law: $\text{Rate} = k$	Rate Law: $\text{Rate} = k[A]$
Integrated Rate Equation: $[A]_t = -kt + [A]_0$	Integrated Rate Equation: $\ln[A]_t = -kt + \ln[A]_0$
Half-Life ($t_{1/2}$): $t_{1/2} = \frac{[A]_0}{2k}$	Half-Life ($t_{1/2}$): $t_{1/2} = \frac{0.693}{k}$
	
-	Useful Numerical Formula: $t = \frac{2.303}{k} \cdot \log\left(\frac{[A]_0}{[A]_t}\right)$

MOLECULARITY VS. ORDER: A KEY DISTINCTION

Molecularity



Definition: Number of reactant species colliding in an elementary step.

Determination: Theoretical, based on the reaction mechanism.

Possible Values: Positive integers (1, 2, rarely 3). Cannot be zero or fractional.

Applicability: Defined only for elementary (single-step) reactions.

Molecularity Limit: Molecularity is almost never greater than 3, as the probability of a simultaneous, effective collision between more than three particles is extremely low.

Order of Reaction



Definition: Sum of powers of concentration terms in the rate law.

Determination: Experimental, must be measured.

Possible Values: Can be zero, positive, negative, or fractional.

Applicability: Defined for both elementary and complex reactions.

Slowest Step is Key: In a multi-step reaction, the overall rate is determined by the slowest step, known as the rate-determining step.